PCBC H Parametric & Sequences Test **Name**_____ _____/43

SHOW ALL WORK FOR FULL CREDIT. ***Non-Calculator*** **You have the entire period.**

Honor Code (copy below & sign): On my honor, before God, I have neither given, nor received any aid on this test.

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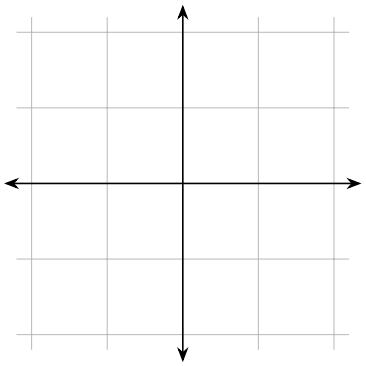
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Signature

Write the pair of parametric equations in rectangular form, including the domain. Then sketch the curve, including direction. (3 points each)

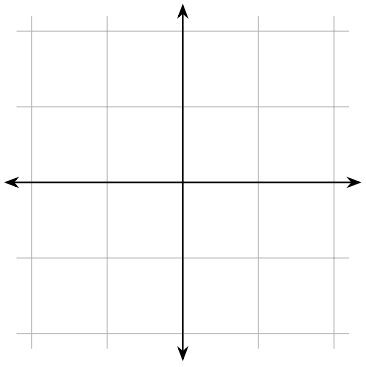
1. $x = 2t + 2, \quad y = t^2 + 2t + 1, \quad -3 \leq t \leq 1$

SIMPLIFIED rectangular form: _____



2. $x = \sqrt{t}, \quad y = 2t - 4, \quad 0 \leq t \leq 4$

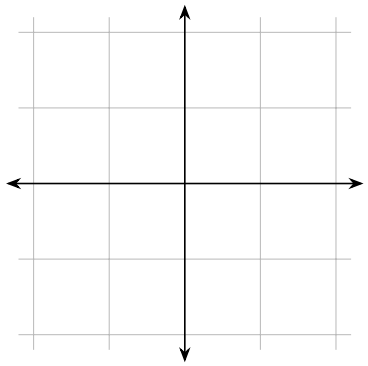
SIMPLIFIED rectangular form: _____



Write the pair of parametric equations in rectangular form. Then sketch the curve. (3 points each)

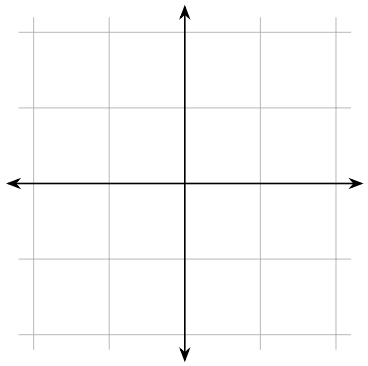
3. $x = 3\cos t, \quad y = 5\sin t, \quad 0 \le t \le 2\pi$

rectangular form: _____



4. $x = 2\sec t, \quad y = 3\tan t$

rectangular form: _____



Convert the rectangular equation to parametric form. Give two different sets of answers. (2 points)

5. $y = -3(x+2)^2 + 5$

The given pattern continues. Write down the nth term of the sequence $\{a_n\}$ suggested by the pattern. (2 points)

6. $\frac{1}{3}, -\frac{2}{9}, \frac{3}{27}, -\frac{4}{81}, \ldots$

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Calculator

Solve the following application problem. Report answers to the thousandths. Include units in your answers. (4 points)

7. A football is kicked from a height of 2 feet at an angle of 38° with an initial speed of 64 ft/s.

a) Write a set of parametric equations for the motion of the football. (Recall:
$x = (v\cos\theta)t,\ \ y = (v\sin\theta)t - 16t^2 + h$)

b) Determine the maximum height reached by the football.

c) Determine how long the football is in the air.

d) Determine how far the football traveled in the air.

Write the first five terms of the sequence. (3 points)

8. $\{a_n\} = \left\{(-1)^{n+1}\dfrac{2^n}{n}\right\}$

A sequence is defined recursively. Write the first five terms. (3 points)

9. $a_1 = 4; \quad a_n = \dfrac{a_{n-1}}{n}$

Write out the sum. (3 points)

10. $\displaystyle\sum_{k=1}^{n}\left(k^2+1\right)$

Express the sum using summation notation. (3 points)

11. $\dfrac{1}{3}+\dfrac{2}{4}+\dfrac{3}{5}+\cdots+\dfrac{15}{17}$

Find each sum. Show the properties and formulas you use, then use your calculator. (#12 is 3 points and #13 is 4 points)

12. $\sum_{k=1}^{30} \left(2k^2 - 5k\right)$

13. $\sum_{k=9}^{24} 3k^3$

Expand using the Binomial Theorem. (4 points)

14. $(2x-5)^4$

Answer Key

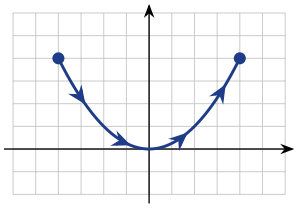
Work on the left, the reason beside it. Boxed = final answer.

Day 1: Non-Calculator

1. $x = 2t+2,\ y = t^2+2t+1,\ -3 \le t \le 1$

t	x	y
-3	-4	4
-2	-2	1
-1	0	0
0	2	1
1	4	4

brackets: plug in both endpoints



solid dots, no end arrows: closed interval

$x-2 = 2t \ \Rightarrow\ t = \frac{x-2}{2}$ — *isolate t in x: one answer, no $\pm$*

$y = \left(\frac{x-2}{2}\right)^2 + 2\left(\frac{x-2}{2}\right) + 1$ — *substitute*

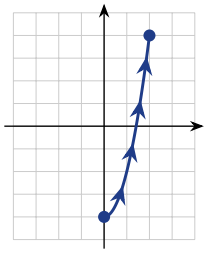
$y = \frac{x^2-4x+4}{4} + x - 2 + 1$

$\boxed{y = \frac{1}{4}x^2 \ \text{ for } x \text{ in } [-4,4]}$ — *SIMPLIFIED = y alone; domain = x from the t-chart*

Other way: $y = (t+1)^2$ and $t+1 = \frac{x}{2}$, so $y = \frac{x^2}{4}$.

2. $x = \sqrt{t},\ y = 2t-4,\ 0 \le t \le 4$

t	x	y
0	0	-4
1	1	-2
4	2	4



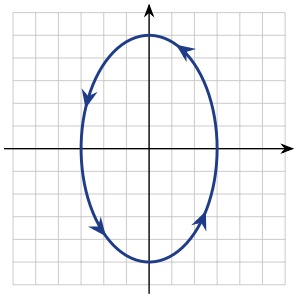
$x^2 = t$ — *square both sides: one answer for t*

$y = 2x^2 - 4$ — *substitute*

$\boxed{y = 2x^2 - 4 \ \text{ for } x \text{ in } [0,2]}$ — *$x = \sqrt{t}$ is never negative, so only the right half*

3. $x = 3\cos t,\ y = 5\sin t,\ 0 \le t \le 2\pi$

t	(x,y)
0	$(3,0)$
$\frac{\pi}{2}$	$(0,5)$
π	$(-3,0)$
$\frac{3\pi}{2}$	$(0,-5)$
2π	$(3,0)$



counterclockwise from $(3,0)$

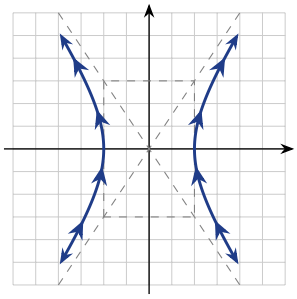
$\cos t = \frac{x}{3}, \quad \sin t = \frac{y}{5}$ — *solve each for the trig function*

$\sin^2 t + \cos^2 t = 1 \Rightarrow \left(\frac{y}{5}\right)^2 + \left(\frac{x}{3}\right)^2 = 1$ — *identity*

$\boxed{\frac{x^2}{9} + \frac{y^2}{25} = 1}$ — *whole ellipse is traced: no restriction*

4. $x = 2\sec t,\ y = 3\tan t$ (no interval given)

t	(x,y)
$-\frac{\pi}{4}$	$(2\sqrt{2},-3)$
0	$(2,0)$
$\frac{\pi}{4}$	$(2\sqrt{2},3)$
$\frac{3\pi}{4}$	$(-2\sqrt{2},-3)$
π	$(-2,0)$
$\frac{5\pi}{4}$	$(-2\sqrt{2},3)$



box from $a = 2$, $b = 3$; asymptotes through the corners; both branches run bottom to top

$\sec t = \frac{x}{2}, \quad \tan t = \frac{y}{3}$

$1 + \tan^2 t = \sec^2 t \Rightarrow 1 + \frac{y^2}{9} = \frac{x^2}{4}$ *identity*

$\boxed{\frac{x^2}{4} - \frac{y^2}{9} = 1}$ *t-chart: $(2,0)$ and $(-2,0)$, both branches: whole graph, done*

also right: $y = \pm 3\sqrt{\frac{x^2}{4} - 1}$ *keep $\pm$; $+\sqrt{}$ alone is wrong*

5. $y = -3(x+2)^2 + 5$

$x = t,\ y = -3(t+2)^2 + 5$ *replace x with t*

$\boxed{x = t,\ y = -3(t+2)^2 + 5,\ t \text{ in } (-\infty,\infty)}$

$x = t - 2:\ y = -3(t-2+2)^2 + 5$ *pick $t-2$ so the inside becomes t*

$\boxed{x = t - 2,\ y = -3t^2 + 5,\ t \text{ in } (-\infty,\infty)}$

Give both $x =$ and $y =$ ($t = x+2$ is not parametric form). "For t in $(-\infty,\infty)$" is assumed for a whole parabola; writing it is fine. Don't expand $-3(t+2)^2+5$; it's not asked and it wastes time.

6. $\overset{1}{\frac{1}{3}},\ \overset{2}{-\frac{2}{9}},\ \overset{3}{\frac{3}{27}},\ \overset{4}{-\frac{4}{81}}$

top $= n$, bottom $= 3^n$

signs $+,-,+,-\ \Rightarrow (-1)^{n+1}$ *first term positive, so $n+1$ ($(-1)^n$ makes it negative)*

$\boxed{a_n = (-1)^{n+1}\frac{n}{3^n}}$ *braces { } also fine*

Day 2: Calculator

7. Football: $h = 2$, $\theta = 38°$, $v = 64$ ft/s. Calculator in **degree** mode, Graph in **Parametric**.

a) $x = (64\cos 38°)t$ — *plug v and θ into the given formula*

$y = (64\sin 38°)t - 16t^2 + 2$ — *+h: kicked from 2 ft, so +2*

$x_1(t) = 64\cos(38°)\cdot t,\ \ y_1(t) = 64\sin(38°)\cdot t - 16t^2 + 2$ — *set up: what you type in*

$0 \le t \le 3$, tstep 0.01; x : 0 to 140, y : 0 to 30 — *if the graph doesn't come back down, raise tmax*

b) $\boxed{y \approx 26.258 \text{ ft}}$ — *trace to the top of the path ($t \approx 1.231$)*

c) $\boxed{t \approx 2.512 \text{ s}}$ — *trace to where $y \approx 0$; trace step 0.01, then 0.001*

d) $\boxed{x \approx 126.707 \text{ ft}}$ — *same point, read x. trace landing on $t = 2.512$ shows ≈ 126.687; a trace answer like that is fine*

*Other way for c): $0 = -16t^2 + (64\sin 38°)t + 2$, quadratic formula, keep the positive t. Watch: answers in **feet**, the speed was ft/s.*

8. $(-1)^2\frac{2}{1},\ (-1)^3\frac{4}{2},\ (-1)^4\frac{8}{3},\ (-1)^5\frac{16}{4},\ (-1)^6\frac{32}{5}$

$\boxed{2,\ -2,\ \frac{8}{3},\ -4,\ \frac{32}{5}}$

reduce: $\frac{4}{2} = 2$, $\frac{16}{4} = 4$

9.

$a_1 = 4,\ \ a_2 = \frac{a_1}{2} = \frac{4}{2} = 2$ — *divide by the term number*

$a_3 = \frac{2}{3},\ \ a_4 = \frac{2/3}{4} = \frac{1}{6},\ \ a_5 = \frac{1/6}{5} = \frac{1}{30}$

$\boxed{4,\ 2,\ \frac{2}{3},\ \frac{1}{6},\ \frac{1}{30}}$

10. $(1^2+1) + (2^2+1) + (3^2+1) + \cdots + (n^2+1) = \boxed{2 + 5 + 10 + \cdots + (n^2+1)}$

start at the lower index, write a few terms, ellipsis, end with the upper index n. Don't add them.

11. $\boxed{\displaystyle\sum_{k=1}^{15} \frac{k}{k+2}}$

top counts 1 to 15; bottom is 2 more. Check $k = 1$ gives $\frac{1}{3}$. Same letter top, bottom and inside.

12.

$\displaystyle\sum_{k=1}^{30} 2k^2 - \sum_{k=1}^{30} 5k$ — *(3) split the difference*

$\displaystyle 2\sum_{k=1}^{30} k^2 - 5\sum_{k=1}^{30} k$ — *(1) constants out*

$\displaystyle 2\left[\frac{30(31)(61)}{6}\right] - 5\left[\frac{30(31)}{2}\right]$ — *(7), (6) with $n = 30$. Stop here, then calculator*

$\boxed{16585}$

13.

$\displaystyle 3\sum_{k=9}^{24} k^3$ — *(1) constant out*

$\displaystyle 3\left[\sum_{k=1}^{24} k^3 - \sum_{k=1}^{8} k^3\right]$ — *(4) lower index is 9, not 1: $j = 8$*

$\displaystyle 3\left[\left(\frac{24(25)}{2}\right)^2 - \left(\frac{8(9)}{2}\right)^2\right]$ — *(8) whole bracket squared*

$\boxed{266112}$

14. $(2x-5)^4$: $n=4$, row 1 4 6 4 1. $x \to 2x$ (whole thing to the power), $a \to -5$ (the sign goes with it).

$1(2x)^4(-5)^0 + 4(2x)^3(-5)^1 + 6(2x)^2(-5)^2$ *blanks first: 2x powers count down, −5 powers count up*

$+4(2x)^1(-5)^3 + 1(2x)^0(-5)^4$

$1(16x^4) + 4(8x^3)(-5) + 6(4x^2)(25) + 4(2x)(-125) + 625$ *odd power of −5 is negative*

$\boxed{16x^4 - 160x^3 + 600x^2 - 1000x + 625}$